



SAMUEL J. TUSICK

Pittsburgh, PA (Ambridge) | Fort Myers, FL

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PROFESSIONAL SUMMARY

Software Engineering student (4.0 in-major GPA) at Florida Gulf Coast University building AI-powered, cloud-native, full-stack applications. Student Research Assistant at the FGCU Dendritic AI & DS Institute, contributing to faculty-supervised research on LLM-integrated tutoring systems and a serverless AWS virtual-patient training platform. Prior experience spans early-stage AI startup development, university-level research, and technical support. Experienced leading development teams and delivering software in collaboration with faculty, researchers, and cross-functional stakeholders.

EDUCATION

Florida Gulf Coast University — B.S., Software Engineering

Expected May 2027

Fort Myers, FL

- In-Major GPA: 4.0 | Dean's List (2023, 2024, 2025)
- Honors: Ruth M. and Cleveland L. Campbell Engineering Scholarship Endowed Fund; Kunkle Family Scholarship Endowed Fund
- Relevant Coursework: Machine Learning, AI Agent Studio, Software Architecture & Design, Intro to Data Engineering, Software Testing, Requirements Engineering & Analysis, Intro to Cloud Computing, Data Structures & Algorithms, Software Engineering Fundamentals, Assembly Language Programming, Computer Security, Operating Systems, Calculus III, Differential Equations

Ambridge Area High School — High School Diploma

August 2019 – May 2023

Ambridge, PA

TECHNICAL SKILLS

- Languages: C, C++, Python, JavaScript, SQL, Java
- Applied AI: Agentic AI system design, LLM-integrated tutoring systems, prompt-driven application design
- Cloud: AWS (API Gateway, Lambda, Aurora, Cognito, IAM), Oracle Cloud Infrastructure
- Backend: Flask, PostgreSQL, Supabase, SQLite
- Frontend: React, Vite, Tailwind CSS
- Testing & Dev Tools: Selenium, Automated Software Testing, Git, GitHub Actions, Jira, Trello, pgAdmin, VS Code, CLion

PROFESSIONAL EXPERIENCE

FGCU Dendritic AI & DS Institute — Student Research Assistant

March 2025 – Present

Fort Myers, FL | On-site

- Designed, developed, and maintain 14+ custom AI tutors actively used within the Engineering Learning Hub to support tutoring, mentoring, and knowledge assistance.
- Supported the coordination and delivery of the 2026 FGCU Summer AI Academy, serving 500+ participants through workshops, trainings, webinars, and 150+ hours of instructional content.
- Collaborate with faculty and researchers to create applied AI learning experiences for students, educators, and community organizations.
- Contribute as a developer on a master's thesis project building an AI Avatar Training System for mental health training scenarios.
- Research and evaluate emerging Agentic AI frameworks and technologies for educational and operational deployments.

Authentifi.AI — Platform Developer

January 2026 – March 2026

Naples, FL

- Participated in software development discussions within an early-stage AI startup environment.
- Worked within an Agile development environment, using Trello for task management, sprint planning, and team collaboration.

- Updated previously hard-coded system data points to retrieve real-time information through API integrations connected to backend databases.
- Assisted with exploratory development efforts related to AI-enabled platform technologies, gaining exposure to product development workflows and iterative delivery practices.

Eagle CyberNest Lab — Router-Level Security Detection, Subteam Lead & Backend Developer February 2025 – April 2025
Fort Myers, FL | On-site

- Led a five-member software engineering team responsible for backend architecture, task coordination, technical decision-making, and milestone delivery on a home-network security research platform.
- Built Flask backend services integrating three router platforms, using SSH and SQLite to enable secure, real-time monitoring and centralized data collection across multiple network devices.
- Tested and deployed production-ready router-communication scripts, improving system reliability and security.

Princeton University — Research Assistant, Computational Usage in Different Fields of Study February 2023 – June 2023
Princeton, NJ

- Supported a research team studying how programming is applied across academic and professional fields.
- Provided transcription support for interviews with 300+ professors and professionals on how they integrate programming into their research and work.
- Collected, organized, and documented interview data to ensure accessibility for analysis.

Ambridge Area High School — Tech Intern January 2023 – June 2023
Ambridge, PA

- Provided technical support to students and staff, troubleshooting network issues, user account problems, and display difficulties.
- Assisted with desk configurations for staff workstations, ensuring comfort, accessibility, and functionality.
- Supported setup of school events and assemblies, including projector and audio equipment.

PROJECTS

MAIVA – AI Virtual Patient Training Platform In Progress
React, Vite, Tailwind CSS, AWS (API Gateway, Lambda, Aurora, Cognito, IAM)

- Co-designing system architecture and requirements for an AI-powered virtual patient platform supporting graduate mental health education, translating stakeholder input from two university colleges into structured data models for realistic, LLM-driven patient personas.
- Building the React, Vite, and Tailwind CSS frontend and integrating it with a serverless AWS backend.
- Scoping Unreal Engine-based avatar rendering as the platform's next phase of development.

Switch4Good Database System January 2026 – May 2026
PostgreSQL, SQL, Data Engineering

- Led development of a PostgreSQL database supporting nonprofit operations across 100+ universities and 1,500+ students.
- Designed a PostgreSQL schema to manage student participation, projects, institutional partnerships, and program outcomes.

NHL Stat Scraper
Python, Flask, Selenium, React, Vite, Render, Netlify

- Built a full-stack analytics application allowing users to search and compare live NHL player statistics.
- Built a Flask backend using Selenium to scrape data from the official league site and expose it through REST APIs.
- Developed a React frontend with Vite enabling users to filter and dynamically view stat categories via query parameters.

Pinterest Testing Suites March 2026 – May 2026
Java, Selenium

- Led a software testing team using Selenium and Agile testing methodologies to evaluate functionality, usability, and regression scenarios.
- Developed Selenium-based test suites in Java and presented findings on functionality, reliability, and usability.

LEADERSHIP & INVOLVEMENT

- FGCU AI Day — Panelist (March 2026): Presented on "AI in University Operations," showcasing RAG-enhanced AI tutors built for the Engineering Learning Hub.
- Pinterest Testing Suites — Team Lead (March 2026 – May 2026)
- Switch4Good Database System — Team Lead (January 2026 – May 2026)
- Eagle CyberNest Lab — Subteam Lead, Router-Level Security Detection, 5-member team (February 2025 – April 2025)
- Deeptech Club — Member (June 2026 – Present)
- Computer Science & Software Engineering Club — Member (August 2024 – Present)

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate